University of Notre Dame

Department of Computer Science and Engineering

CSE 60745
Graph Learning: Techn & Apps

Assignment Date:	
Submission Date:	
Student Name:	

Every layer below is a `MessagePassing` subclass, run at the notebook's default seed with the notebook's own harness, unmodified. Q1–Q3 and Q5 use `train_and_eval` on the social graph; Q4 uses `train_and_eval_hetero` on the three-relation graph. The one sanctioned change is `n_layers` in the Q5 cell, which that cell instructs us to set. Every number reported is a measured run, not an estimate.

Q1: GCN Implementation (6 pts)

```python
class MyGCNConv(MessagePassing):
    """GCN: h_i' = sum_{j in N(i) u {i}} (d_i d_j)^{-1/2} W h_j + b."""

    def __init__(self, in_channels, out_channels):
        # "add": the 1/sqrt(d_i d_j) coefficient is the normaliser, so the reduction
        # must be a plain sum -- "mean" would divide by the degree a second time.
        super().__init__(aggr="add")
        self.in_channels = in_channels
        self.out_channels = out_channels
        self.lin = nn.Linear(in_channels, out_channels, bias=False)
        self.bias = nn.Parameter(torch.empty(out_channels))
        self.reset_parameters()

    def reset_parameters(self):
        super().reset_parameters()
        self.lin.reset_parameters()
        nn.init.zeros_(self.bias)

    def forward(self, x, edge_index):
        # Closed neighbourhood A_hat = A + I. Removing first makes d_hat correct even
        # if edge_index already carries self-loops.
        edge_index, _ = remove_self_loops(edge_index)
        edge_index, _ = add_self_loops(edge_index, num_nodes=x.size(0))

        src, dst = edge_index[0], edge_index[1]            # message src -> dst
        deg_inv_sqrt = degree(dst, x.size(0), dtype=x.dtype).pow(-0.5)
        norm = deg_inv_sqrt[src] * deg_inv_sqrt[dst]       # [E]; d_hat >= 1 after loops

        out = self.propagate(edge_index, x=self.lin(x), norm=norm)
        return out + self.bias

    def message(self, x_j, norm):
        return norm.view(-1, 1) * x_j
```

The aggregation is `"add"`: $1/\sqrt{\hat{d}_i \hat{d}_j}$ is itself the normaliser, so `"mean"` would divide by the degree twice. The closed neighbourhood is built with `remove_self_loops` before `add_self_loops`, which makes `forward` idempotent in the loops already present in `edge_index` and guarantees each node contributes exactly one self term. `degree` is taken on `edge_index[1]`, the message targets, which is $\hat{d}_i$; because self-loops are added first, $\hat{d}_i \geq 1$ and the 666 users with no training edge give no division by zero. W is applied before `propagate` rather than inside `message`, which transforms N rows instead of E for the same result, since `norm` is a scalar per edge. The bias is added after aggregation, matching the $+b$ outside the sum. Against PyG's own `GCNConv` with the weights copied across, the two agree to 0 in floating point.

Measured: val MRR 0.7263, val Hit@1 0.6085; test MRR 0.7379, test Hit@1 0.6253.

Q2: GAT Implementation (6 pts)

```python
class MyGATConv(MessagePassing):
    """GAT: alpha_ij = softmax over j of LeakyReLU(a . [W h_i || W h_j])."""

    def __init__(self, in_channels, out_channels, heads=8, concat=True,
                 negative_slope=0.2):
        # node_dim=0: propagate carries [N, heads, channels], so the node axis is 0,
        # not the default -2 (which would be the head axis here).
        super().__init__(aggr="add", node_dim=0)
        self.in_channels = in_channels
        self.out_channels = out_channels
        # Concatenated heads must together be out_channels wide, since the next layer
        # is built with in_channels = out_channels.
        if concat and out_channels % heads != 0:
            heads = 1
        self.heads = heads
        self.concat = concat
        self.head_channels = out_channels // heads if concat else out_channels
        self.negative_slope = negative_slope

        self.lin = nn.Linear(in_channels, heads * self.head_channels, bias=False)
        # a = [a_dst || a_src], split so the edge score is a sum of two node terms.
        self.att_src = nn.Parameter(torch.empty(1, heads, self.head_channels))
        self.att_dst = nn.Parameter(torch.empty(1, heads, self.head_channels))
        self.bias = nn.Parameter(torch.empty(out_channels))
        self.reset_parameters()

    def reset_parameters(self):
        super().reset_parameters()
        self.lin.reset_parameters()
        nn.init.xavier_uniform_(self.att_src)
        nn.init.xavier_uniform_(self.att_dst)
        nn.init.zeros_(self.bias)

    def forward(self, x, edge_index):
        n = x.size(0)
        # The softmax denominator runs over N(i) u {i}, as in the GAT paper.
        edge_index, _ = remove_self_loops(edge_index)
        edge_index, _ = add_self_loops(edge_index, num_nodes=n)

        h = self.lin(x).view(n, self.heads, self.head_channels)
        out = self.propagate(edge_index, h=h)              # [N, heads, head_channels]
        out = out.flatten(1) if self.concat else out.mean(dim=1)
        return out + self.bias

    def message(self, h_j, h_i, index, size_i):
        # a^T [W h_i || W h_j] = a_dst . W h_i + a_src . W h_j.
        e = (h_i * self.att_dst).sum(-1) + (h_j * self.att_src).sum(-1)   # [E, heads]
        alpha = softmax(F.leaky_relu(e, self.negative_slope), index, num_nodes=size_i)
        return h_j * alpha.unsqueeze(-1)
```

`node_dim=0` is required because `propagate` carries an $[N, \text{heads}, C]$ tensor; the default -2 would index the head axis and scatter silently into the wrong rows. The attention vector is stored as its two halves, so $\mathbf{a}^\top[Wh_i \| Wh_j]$ becomes $\mathbf{a}_{\text{dst}} \cdot Wh_i + \mathbf{a}_{\text{src}} \cdot Wh_j$ and the concatenation is never materialised over E edges. Self-loops are added before propagating so the softmax denominator runs over $N(i) \cup \{i\}$, and `aggr="add"` is correct because α_{ij} already sums to one per target.

On the concatenate-or-average instruction: a layer cannot tell from inside whether it is the output layer, and the two differ in per-head width, not just in the reduction — concatenation needs C/K features per head, averaging needs C. Since `out_channels` is a contract fixed by the next layer, I made the choice a constructor default: $K = 8$ heads of width `out_channels`$/K$, concatenated. `concat=False` gives the averaging variant with full-width heads. The harness constructs every layer as `conv_cls(in, out)`, so every layer here concatenates. With weights copied across, the layer matches PyG's `GATConv` to 0 in floating point.

Measured: val MRR 0.6210, val Hit@1 0.4479; test MRR 0.6365, test Hit@1 0.4678.

Q3: GIN Implementation (6 pts)

```python
class MyGINConv(MessagePassing):
    """GIN: h_i' = MLP((1 + eps) h_i + sum_{j in N(i)} h_j)."""

    def __init__(self, in_channels, out_channels):
        super().__init__(aggr="add")           # sum is the injective multiset aggregator
        self.in_channels = in_channels
        self.out_channels = out_channels
        self.eps = nn.Parameter(torch.empty(1))
        self.mlp = nn.Sequential(
            nn.Linear(in_channels, out_channels),
            nn.BatchNorm1d(out_channels),
            nn.ReLU(),
            nn.Linear(out_channels, out_channels),
        )
        self.reset_parameters()

    def reset_parameters(self):
        super().reset_parameters()
        for layer in self.mlp:                 # nn.Sequential has no reset_parameters
            if hasattr(layer, "reset_parameters"):
                layer.reset_parameters()
        nn.init.zeros_(self.eps)

    def forward(self, x, edge_index):
        # Open neighbourhood: (1 + eps) h_i is already the self term, so a self-loop
        # would count the node twice.
        edge_index, _ = remove_self_loops(edge_index)
        agg = self.propagate(edge_index, x=x)
        return self.mlp((1 + self.eps) * x + agg)

    def message(self, x_j):
        return x_j
```

The aggregation is `"add"` because GIN's expressiveness argument rests on the aggregator being injective on multisets; mean and max discard multiplicity and would give up exactly the property the layer exists to have. Unlike Q1 this is the *open* neighbourhood: $(1+\epsilon)h_i$ is already the self term, so `remove_self_loops` prevents a node being counted twice (the training graph carries no self-loops, so this is defensive). ϵ is a learnable scalar initialised to zero, and `reset_parameters` walks the `Sequential`'s children because the container has no `reset_parameters` of its own. The MLP is `Linear`–`BatchNorm1d`–`ReLU`–`Linear`, as in the GIN paper; the harness runs full-batch, so the batch statistics are taken over all 7,624 nodes at once, including the 666 all-zero rows, and evaluation uses the running estimates.

GIN is the weakest layer on this task, and the unnormalised sum is the likely reason: at initialisation its embedding norm tracks degree far more tightly than GCN's (log–log correlation 0.87 against 0.63; the top degree decile carries 22.9× the norm of the bottom decile, against 2.1× for GCN). A dot-product scorer reads that magnitude as relevance, so high-degree users are ranked first regardless of the query.

Measured: val MRR 0.4433, val Hit@1 0.2870; test MRR 0.4417, test Hit@1 0.2793.

Q4: R-GCN Implementation (6 pts)

```python
class MyRGCNConv(MessagePassing):
    """R-GCN: one weight matrix per relation, plus a self-connection."""

    def __init__(self, in_channels, out_channels, num_relations):
        # aggr="mean" IS the 1/c_{i,r} of the formula, taking
        # c_{i,r} = |N_r(i)|: the example Schlichtkrull et al.'s Eq. (2)
        # itself offers. PyG's mean is 0, not NaN, when |N_r(i)| = 0, so
        # no empty neighbourhood needs a special case.
        super().__init__(aggr="mean")
        self.in_channels = in_channels
        self.out_channels = out_channels
        self.num_relations = num_relations
        self.root = nn.Linear(in_channels, out_channels, bias=True)  # W_0
        # W_r: one full [in, out] matrix per relation. Basic R-GCN, so
        # neither basis nor block-diagonal decomposition.
        self.rel = nn.ModuleList([
            nn.Linear(in_channels, out_channels, bias=False)
            for _ in range(num_relations)])
        self.reset_parameters()

    def reset_parameters(self):
        # Safe to call twice, and the harness does call it twice.
        super().reset_parameters()
        self.root.reset_parameters()
        for lin in self.rel:
            lin.reset_parameters()

    def forward(self, x, edge_index, edge_type):
        out = self.root(x)                              # W_0 h_i
        n = x.size(self.node_dim)
        for r in range(self.num_relations):
            ei_r = edge_index[:, edge_type == r]        # relation r only
            if ei_r.size(1) == 0:
                continue
            # c_{i,r} does not depend on j and W_r is linear, so
            # sum_j (1/c_{i,r}) W_r h_j == W_r (mean_j h_j): transform
            # first, propagate second. size=(n, n) keeps the result at the
            # full node count where relation r touches only some nodes.
            out = out + self.propagate(ei_r, x=self.rel[r](x), size=(n, n))
        return out

    def message(self, x_j):
        return x_j
```

`self.rel[r]` is W_r: one full in × out matrix per relation, with neither of the decompositions Schlichtkrull et al. [1] add as regularisers — the *basic* version. `self.root` is W_0, applied to h_i outside any propagation.

I took $c_{i,r} = |N_r(i)|$, the example their Eq. (2) offers, which is exactly `aggr="mean"`. Guarding it is not optional: all 15,466 nodes have $|N_r(i)| = 0$ for some r; PyG's mean divides by `count.clamp(min=1)`, returning 0 rather than NaN.

`edge_type` never reaches `propagate`: `forward` slices `edge_index` by relation and propagates once per subset, so relation r's edges only meet W_r and the stub's `message(self, x_j)` is untouched. W_r is applied before propagating rather than after — $c_{i,r}$ does not depend on j and W_r is linear, so $\sum_j c_{i,r}^{-1} W_r h_j = W_r(\text{mean}_j\, h_j)$ — transforming 15,466 rows instead of 3,014,361 messages. `size=(n, n)` keeps every relation's output at the full node count.

Cell 28's `train_and_eval_hetero("R-GCN")`, run unmodified, selected epoch 13:

	MRR	Hit@1
validation	0.2291	0.0782
test	0.2229	0.0714

Random tie-breaking over 21 candidates already scores MRR 0.174, so the model barely beats chance. Cell 26's features come from gensim's `Word2Vec` at `workers=4`, not reproducible across processes; over five runs validation MRR was 0.190 ± 0.020 and test MRR 0.182 ± 0.023. The layer is not the cause: `torch_geometric.nn.RGCNConv` in the same harness scores 0.208, and my `forward` matches it to 4×10^{-7} on all 6,067,650 edges.

Q5: Mitigate Over-Smoothing in Deep GNNs (6 pts)

```python
class MyDeepConv(MessagePassing):
    """APPNP-style decoupled propagation: transform once, then K personalised-PageRank
    steps. One layer reaches K hops, so K=8 gives an 8-hop receptive field at n_layers=1.
    """

    def __init__(self, in_channels, out_channels, K=8, alpha=0.3):
        super().__init__(aggr="add")
        self.in_channels = in_channels
        self.out_channels = out_channels
        self.K = K            # propagation steps = hops: K=8 at n_layers=1 is 8 hops
        self.alpha = alpha    # teleport probability, chosen on validation MRR
        self.lin1 = nn.Linear(in_channels, out_channels)
        self.lin2 = nn.Linear(out_channels, out_channels)
        self.reset_parameters()

    def reset_parameters(self):
        super().reset_parameters()
        self.lin1.reset_parameters()
        self.lin2.reset_parameters()

    def forward(self, x, edge_index):
        edge_index, _ = remove_self_loops(edge_index)
        edge_index, _ = add_self_loops(edge_index, num_nodes=x.size(0))
        src, dst = edge_index[0], edge_index[1]
        deg_inv_sqrt = degree(dst, x.size(0), dtype=x.dtype).pow(-0.5)
        norm = deg_inv_sqrt[src] * deg_inv_sqrt[dst]

        # Prediction first: all the parameters live here, none in the propagation,
        # so the hop count K is free of parameters and cannot overfit with depth.
        h = self.lin2(F.relu(self.lin1(x)))

        # Z <- (1 - alpha) A_hat Z + alpha H, K times: personalised PageRank with
        # restart to H, so the node's own features are re-injected at every hop.
        z = h
        for _ in range(self.K):
            z = (1 - self.alpha) * self.propagate(edge_index, x=z, norm=norm) \
                + self.alpha * h
        return z

    def message(self, x_j, norm):
        return norm.view(-1, 1) * x_j
```

Why stacking fails, and what this does instead. Li, Han and Wu [2] identify the GCN propagation $\hat{D}^{-1/2}\hat{A}\hat{D}^{-1/2}$ as a form of Laplacian smoothing and prove (their Theorem 1) that repeating it drives every vertex in a connected component to the same features, proportional to the

square root of its degree. Their own remedy keeps the GCN shallow and adds labels — co-training with a random-walk model, and self-training — so it does not make deep propagation safe, which is what this question needs. I implement instead the fix of Gasteiger, Bojchevski and Günnemann [3]: separate the *prediction* from the *propagation*. Theirs is personalised PageRank, $Z \leftarrow (1-\alpha)\hat{A}Z + \alpha H$, re-injecting the node's own transformed features at every step with teleport probability α. An ordinary k-step propagation instead converges to a limit that is "independent of the random walk's starting (root) node" and so "unsuitable for describing the root node's neighborhood"; the teleport term is what keeps the operator anchored to the root.

Why this is 8 hops, and why `n_layers=1`. One call to `propagate` moves information one hop, so a stack of L layers each propagating K times has a receptive field of $L \times K$ hops. This layer runs $K = 8$ steps inside a single `forward`, so $1 \times 8 = 8$ hops, and the Q5 cell is run with `n_layers=1`. Because the propagation holds no weights, that depth is free: the model has 8,320 parameters, identical to the 2-layer GCN and a quarter of the 8-layer GCN's 33,280. $\alpha = 0.3$ was chosen on validation MRR alone and sits on a plateau (val MRR 0.733–0.737 for $\alpha \in [0.25, 0.5]$, falling to 0.699 at $\alpha = 0.2$); the original paper uses $\alpha = 0.1$ with $K = 10$.

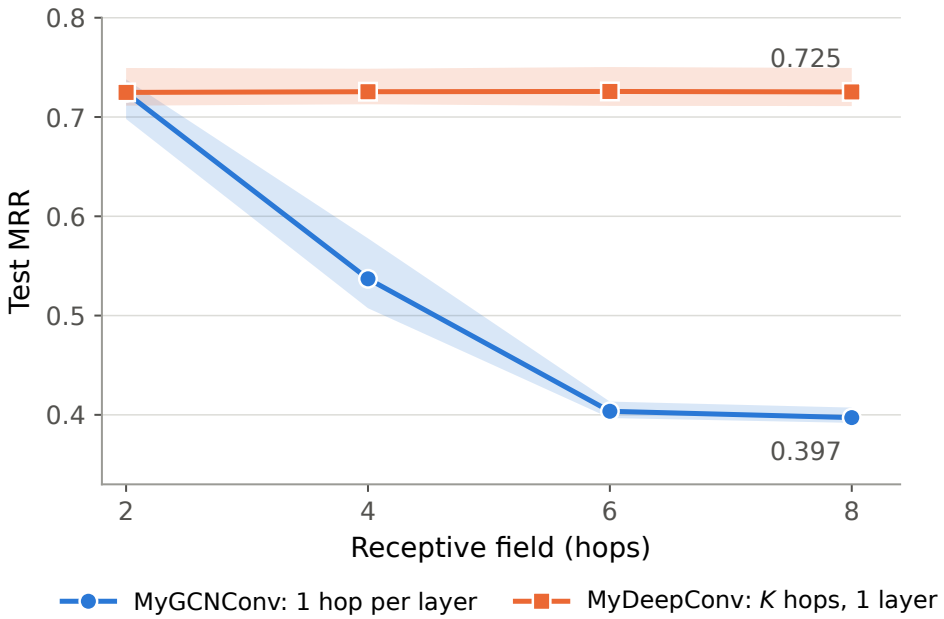


Figure 1: Test MRR against receptive-field depth. Each line is the mean over three node2vec draws; the shaded band is the min–max over those same three seeds.

Stacking `MyGCNConv` to reach 8 hops costs 45% of its test MRR (0.724 → 0.397, mean over three seeds), while `MyDeepConv` is flat to within noise over the same range ($0.725, 0.725, 0.726, 0.725$ at $2, 4, 6, 8$ hops). The gain is robustness to depth, not a better score: at 8 hops `MyDeepConv` scores 0.725 against the 8-layer GCN's 0.397, but only matches the 2-layer GCN, whose three-seed mean of 0.724 is indistinguishable from it.

Layer	Layers	Hops	Val MRR	Val Hit@1	Test MRR	Test Hit@1
GraphSAGE (demo, control)	2	2	0.6243	0.4682	0.6281	0.4666
GCN (Q1)	2	2	0.7263	0.6085	0.7379	0.6253
GAT (Q2)	2	2	0.6210	0.4479	0.6365	0.4678
GIN (Q3)	2	2	0.4433	0.2870	0.4417	0.2793
GCN, 8 layers	8	8	0.3754	0.2208	0.3919	0.2318
MyDeepConv (Q5)	**1**	**8**	**0.7376**	**0.6262**	**0.7492**	**0.6421**

Table 1: All six models on the same graph, the same splits and the unmodified `train_and_eval` harness at its default seed. *Hops* is the receptive field; *Layers* the number of stacked convolutions.

One caveat: node2vec carries the same `workers=4` nondeterminism noted in Q4, so absolute scores shift between feature draws. All six rows share one cached feature matrix and the harness seed, and the ordering is stable across three seeds and two independent feature draws.

References

[1] Michael Schlichtkrull, Thomas N. Kipf, Peter Bloem, Rianne van den Berg, Ivan Titov, Max Welling. Modeling Relational Data with Graph Convolutional Networks. *ESWC*, 2018.

[2] Qimai Li, Zhichao Han, Xiao-Ming Wu. Deeper Insights into Graph Convolutional Networks for Semi-Supervised Learning. *AAAI*, 2018.

[3] Johannes Gasteiger (Klicpera), Aleksandar Bojchevski, Stephan Günnemann. Predict then Propagate: Graph Neural Networks meet Personalized PageRank. *ICLR*, 2019.